

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
First Semester 2025-2026
Mid-Semester Test
(EC-2 Make Up)

Course No. : AIMLCZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 03.01.2026 _FN

No. of Pages	= 4
No. of Questions	= 6

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Answer all the questions:

1. You are a Data Intern at Vertex Auto Parts, reviewing 10-day production output (units/day) post-automation:
190, 250, 240, 270, 230, 280, 245, 265, 285, 380
Provide the five-number summary and identify outliers [3 marks]
Based on the distribution (especially the spread and extreme values), what does this suggest about the stability and consistency of the production process? Is the variation likely due to routine fluctuations, or does it indicate a process shift or data quality concern? Justify briefly. [2 marks]
2. The probabilities that an academican, a businessman, or a politician is appointed Vice-Chancellor of a university are 0.5, 0.3 and 0.2 respectively. The probabilities that research work is promoted given the appointment of each are:
Academican: 0.8, Businessman: 0.6, Politician: 0.4. It is observed that research work has been promoted. Find the probability that the Vice-Chancellor is an academican. [5 marks]
3. The final exam scores for ISM are modeled as normally distributed with mean 73 and standard deviation 8. A teaching assistant makes the following claims:
 - a) Most students scored below 91.
 - b) Scoring between 65 and 89 puts you in the central majority.
 - c) Only 5% of students scored above a certain threshold — that's the cut-off for the Dean's List. Using the model:
 - (i) Verify the first claim: Find $P(X < 91)$. Is it indeed "most"? [2 marks]
 - (ii) Quantify the second claim: Find $P(65 < X < 89)$ [2 marks]
 - (iii) Determine the exact cut-off score for the Dean's List. [1 mark]
4. A cybersecurity operations team models joint behaviour of two discrete metrics during night shifts:

Let X = number of servers compromised (0, 1, 2, 3),

Y = severity level of the attack (values: -1, 0, 1, 2). where (X, Y) has joint probability mass function $f(x, y) = c(x^2 + y + 1)$ for $x \in \{0, 1, 2, 3\}$, $y \in \{-1, 0, 1, 2\}$

(Note: The domain includes negative severity — interpreted as “false alarm” or “low-impact probe.”)

- (a) Find the normalizing constant c [1 mark]
- (b) Compute $P(X = 1, Y \geq 0)$ [1 mark]
- (c) Are X and Y independent? Justify using marginal distributions. [1 mark]
- (d) Given that at least one server was compromised ($X \geq 1$), what is the probability that the attack severity was non-negative ($Y \geq 0$)? [2 marks]

5. A data centre monitors the normalized overheating index X (unitless, range: 0 to 3). The cumulative distribution function (CDF) of X is:

$$F(x) = \begin{cases} 0, & x \leq 0 \\ kx^2, & 0 < x \leq 1 \\ k(2x - 1), & 1 < x \leq 3 \\ 1, & x > 3 \end{cases}$$

$X=0$ means fully cooled; $X=3$ triggers emergency shutdown.

- a) Find the constant k [2 marks]
- b) Find the probability that the system is in warning zone (i.e., $1 \leq x \leq 2$). [2 marks]
- c) Compute the probability density function $f(x)$ [1 mark]

6. a) At Secure Bank, historical data shows that cheque-processing errors (e.g., amount mismatch, signature miss) occur randomly and independently at an average rate of 1.2 errors per 100 cheques processed. On a typical day, the teller processes 250 cheques. Let X = number of errors observed in a day.

- (i) Specify the distribution of X , including its parameter. [1 mark]
- (ii) Find the probability that at most 2 errors occur in a day. [1 mark]
- (iii) What is the probability of observing more than 4 errors? [1 mark]

- b) At Bharat Auto Forge, every brake disc undergoes inspection for three common manufacturing flaws:

W: Warpage (non-flat surface),
 C: Cracking (micro-fractures),
 S: Surface scoring (tool-mark scratches).

A recent audit of 10,000 units revealed: 1,200 had warpage, 850 had cracking, 1,500 had surface scoring, 300 had both warpage and cracking, 400 had both warpage and scoring, 250 had both cracking and scoring, 100 had all three defects

- (i) What is the probability that a randomly selected brake disc has at least one of these three defects? [1 mark]
- (ii) How many discs (out of 10,000) are expected to be completely defect-free? [1 mark]
